React.js Course Syllabus

Description: React.js curriculum from fundamentals through components, hooks, routing, API integration, testing, and deployment.

Module 1: React.js Fundamentals

Module Description: Introduction to React.js, modern frontend development, environment setup, and JSX.

Topic 1: Introduction to React.js

Topic Description:

Lesson 1: What is React.js?

Lesson Description: Understand React and component-based user interfaces.

Lesson 2: React Features

Lesson Description: Learn declarative rendering, reusable components, and the virtual DOM.

Lesson 3: React Application Architecture

Lesson Description: Understand the component tree and one-way data flow.

Topic 2: Environment Setup and JSX

Topic Description:

Lesson 1: Installing Node.js and Vite

Lesson Description: Set up a React project using Node.js, npm, and Vite.

Lesson 2: React Project Structure

Lesson Description: Explore the main files and folders in a React application.

Lesson 3: JSX Syntax

Lesson Description: Write UI markup using JSX expressions and attributes.

Module 2: Components, Props and State

Module Description: Build reusable components and manage interactive user interfaces.

Topic 1: React Components

Topic Description:

Lesson 1: Functional Components

Lesson Description: Create and reuse functional React components.

Lesson 2: Props

Lesson Description: Pass data from parent to child components.

Lesson 3: Conditional Rendering

Lesson Description: Display UI elements based on conditions.

Lesson 4: Rendering Lists and Keys

Lesson Description: Render collections safely with unique keys.

Topic 2: State and Events

Topic Description:

Lesson 1: useState Hook

Lesson Description: Manage local component state.

Lesson 2: Event Handling

Lesson Description: Handle clicks, inputs, and other user interactions.

Lesson 3: Forms and Controlled Inputs

Lesson Description: Build forms with controlled inputs and basic validation.

Module 3: Hooks, Routing and State Management

Module Description: Use React hooks, navigation, and shared application state.

Topic 1: React Hooks

Topic Description:

Lesson 1: useEffect Hook

Lesson Description: Manage effects, subscriptions, and cleanup.

Lesson 2: useRef Hook

Lesson Description: Access mutable values and DOM references.

Lesson 3: useMemo and useCallback

Lesson Description: Understand memoization for selected performance needs.

Lesson 4: Custom Hooks

Lesson Description: Extract reusable stateful logic into custom hooks.

Topic 2: Routing and Shared State

Topic Description:

Lesson 1: React Router

Lesson Description: Configure routes, navigation, and route parameters.

Lesson 2: Context API

Lesson Description: Share state without passing props through every level.

Lesson 3: State Management Basics

Lesson Description: Understand when to use Context or a state management library.

Module 4: API Integration and UI Development

Module Description: Connect React applications to backend APIs and build responsive interfaces.

Topic 1: Working with APIs

Topic Description:

Lesson 1: Fetching Data

Lesson Description: Retrieve server data using fetch or Axios.

Lesson 2: Loading and Error States

Lesson Description: Display loading indicators, empty states, and errors.

Lesson 3: CRUD Operations

Lesson Description: Create, read, update, and delete records through APIs.

Topic 2: Styling and Reusable UI

Topic Description:

Lesson 1: CSS and Responsive Layouts

Lesson Description: Style components using CSS and responsive design techniques.

Lesson 2: Tailwind CSS Basics

Lesson Description: Build utility-first React layouts with Tailwind CSS.

Lesson 3: Reusable UI Components

Lesson Description: Build forms, cards, modals, and tables with reusable patterns.

Lesson 4: Form Validation

Lesson Description: Validate user input and display useful feedback.

Module 5: Advanced React, Testing and Deployment

Module Description: Optimize, test, secure, and deploy a complete React application.

Topic 1: Performance and Quality

Topic Description:

Lesson 1: Lazy Loading and Suspense

Lesson Description: Split application code and load routes on demand.

Lesson 2: React Performance

Lesson Description: Identify unnecessary renders and optimize when needed.

Lesson 3: Component Testing

Lesson Description: Write basic component tests with React Testing Library.

Lesson 4: Error Boundaries

Lesson Description: Handle rendering failures with error boundaries.

Topic 2: Production and Project

Topic Description:

Lesson 1: Authentication and Protected Routes

Lesson Description: Understand login state and route protection in the frontend.

Lesson 2: Environment Variables

Lesson Description: Configure public client-side environment values safely.

Lesson 3: Build and Deployment

Lesson Description: Create production builds and deploy a React application.

Lesson 4: Final React Project

Lesson Description: Develop and deploy a small React application with API integration.